Educational Strategies for Sustainable Futures: Revealing the Priorities and Perspectives of Engineering Students

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Abstract

In 2015, the UN adopted the 17 Sustainable Development Goals (SDGs) as part of the "Agenda 2030". These are interconnected goals that address the main development challenges facing the world. Engineers play an important role in the achievement of the UN's SDGs by developing innovative and sustainable solutions. Therefore, the knowledge, skills, and competencies engineers are equipped with upon graduation are fundamental to be able to actively participate and perform in tackling present and future sustainable problems. Existing studies report the development of self-perceived skills by students after a pedagogical intervention; however, they do not examine why certain skills are prioritized over others. Additionally, most research approaches are quantitative and do not deepen the understanding of why students position themselves concerning sustainability competencies. This study reports on students' perspectives and needs regarding their competence's development for sustainability by examining what they consider as the most important skills to act for sustainability and why. It takes a qualitative approach, where open-ended questions were thematically analyzed. The study involved 18 students from an international master's program in Sustainable Design at Aalborg University, Denmark, and 26 students from the Production Engineering course at the University of Brasília, Brazil. Our results demonstrate a multidimensional vision, prioritizing collaboration, interdisciplinarity, and systemic and problem-oriented thinking. They see the importance of education for sustainability as a motivator for future generations, in addition to the need to know how to look at reality to identify problems and seek a fair and sustainable society. These results point to a holistic and conscious view of students concerning sustainability, and these conclusions can guide educational strategies and initiatives that aim to develop these skills and values among students.

Keywords: Sustainability Education; Competencies Development; Engineering Education.

1 Introduction

Sustainable development seeks to meet the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland et al., 1987), taking into account the balance between the environment, economy, and society (Elkington, 1994). Thus, it is fundamental for an effective transition towards a more ecologically balanced society and economy (EU Science Hub, 2022). In 2015, the UN adopted the 17 Sustainable Development Goals (SDGs) as part of the "Agenda 2030" (UN, 2015). These goals aim to solve global challenges such as poverty, environmental degradation, and climate change and promote peace and prosperity. For these goals to be achieved, a transformative and ecological approach to learning is advocated to enable students to promote sustainability (Boström et al., 2018; Hermes & Rimanoczy, 2018; Sipos et al., 2008).

Competences for Sustainable Development (SD) are now a requirement for individuals and organizations must adopt practices to achieve the UN Sustainable Development Goals (SDGs). According to Wiek (2011), these competences go beyond thematic knowledge and encompass a combination of knowledge, skills, and attitudes to face real sustainability challenges. A distinction has been suggested between basic skills, which address general learning objectives, and key skills for sustainability, which are more specific and detailed (Wiek et al., 2011; Wiek et al., 2015). Wiek (2011) highlights six key competences for sustainability, such as systemic thinking,

anticipatory, normative, strategic thinking, collaboration, and integrated problem solving, which are recognized by UNESCO (2017) as crucial to achieving the SDGs.

Engineers play a key role in promoting a sustainable future through innovation. However, dealing with complex challenges requires specific competences. Studies on the competences necessary for Engineering Education for Sustainable Development (EESD) highlight the importance of problem-solving (Thürer, 2018), interdisciplinary and transdisciplinary skills (Guerra, 2017), strategic skills (Quelhas, 2019), intradisciplinary skills and interpersonal (Ortiz-Marcos, 2020) for educators and engineering professionals. However, there is little research into the competences developed by engineering students in the EESD.

Recent studies on sustainability competences include systematic literature reviews (Annelin & Boström, 2023; Redman & Wiek, 2021) and reports examining the development of students' self-perceived competencies following pedagogical interventions, predominantly using quantitative research methods (Cebrián, Pascual & Moradela, 2019; Cavicchi, 2021; Balakrishnan et al., 2020), which mainly employ Likert-scale self-assessment questionnaires. However, the lack of qualitative approaches makes it difficult to understand student preferences, namely what they consider important and why. Furthermore, the lack of clear criteria for why students choose certain competencies in the questionnaires makes it difficult to correlate them with the key competences for sustainability described by UNESCO (2017).

This study, which is part of a broader research project, aims to explore students' perspectives on the competences needed to act for sustainability and reports on student justifications of why they consider certain competences more important than others.

2 Methodology

2.1 Research context and participants

The study was carried out at the Department of Production Engineering of the Faculty of Technology of the University of Brasília (hereinafter UnB), Brazil, where 26 students from the undergraduate course in Production Engineering participated voluntarily, and at the Department of Sustainability and Planning, Aalborg University (hereinafter AAU), Denmark, where 18 students from an international master's program in Sustainable Design participated in the same way. The research context of the institutions is described in Table 1.

Table 1. Research context of the two educational institutions where the work was carried out.

University of Brasília	Aalborg University
The production engineering course at UnB spans 12 semesters, designed to equip engineers with a systemic approach to tackling engineering challenges. Emphasizing the interaction between professionals and their environments, sustainable development is a key focus. Starting from the 4th semester, students engage in the "Production System Projects" (PSP) course, employing Problem-Based Learning (PBL). Projects undertaken in PSP are offered to external companies pro bono, targeting solutions for customer-raised issues, particularly in sustainable service production systems. Faculty members from UnB's Faculty of Technology guide these projects within the Production Engineering Department.	Since 1974, Aalborg University (AAU) has employed a problem-based, project-organized learning (PBL) model. This approach involves students tackling real-world problems in small groups, with each semester structured around three courses (5 ECTS each) and one problem-based project module (15 ECTS). This method fosters the development of knowledge, competencies, and transversal skills such as problem design, communication, project management, teamwork, and critical thinking. AAU prioritizes the United Nations' Sustainable Development Goals (SDGs), integrating them into its strategic focus. The MSc in Sustainable Design at AAU is a comprehensive engineering program where sustainability is a fundamental value, not an add-on. The program places a strong emphasis on facilitating transitions to sustainability through radical systemic changes.

2.2 Research design

As part of the broader study, participants from UnB and AAU were asked to rank from most important to least important a set of sentences that refer to sustainability competences. The study used Q methodology as the research design (learn more about Q methodology at: https://qmethod.org/), and which results will be reported in two journal articles. Nonetheless, the broader study also comprised open questions, where students are inquired about why they consider certain competences more relevant than others, as well as what type of competences could be further integrated into their educations. Open questions from 39 students were subjected to manual thematic analysis of the data (18 from AAU and 26 from UnB), taking an inductive approach.

The choices were categorized according to the corresponding competence. The justifications were analyzed to understand the reason why that statement was chosen, verifying the consistency of the competence related to the statement. They were then used to draw a qualitative profile of the classes regarding their perspectives on the role of engineers in sustainable development.

3 Results

In both institutions, system thinking, collaboration, and systemic thinking are considered the most important competences; whilst knowledge about SDGs and perceived as least important. The following sub-sections provide an overview of why students consider these three competences more or less important to act for sustainability.

3.1 Engineering complex problem solving and the need for systemic view to understand them

The perception of the engineer's need to have a systemic view of problems and possible solutions is strongly present in students of both courses, as we can see below from the justifications given (for better understanding, the justifications are preceded by a code defined by a letter, where "A" represents AAU students and "B" represents UnB students, followed by a student identification number):

B-7 "*Having a holistic view is essential to develop sustainable solutions that address not only the environmental but the general scope*";

A-8 "*As an engineer, I believe our job is more about analyzing and identifying complex problems, and what are the reasons for unsustainable practices to then take these findings and collaborate with other experts and people that can force change*";

A-14 "*I think that is the base of everything in my profession. It should be almost a way of thinking and working that we have by default without this I don't think other ones make a lot of sense. It should be our culture as sustainable design engineers*";

B-21 *"It is essential to understand the impacts that a project can cause, understanding the economic, social and environmental nature of the decisions that define whether it is executable or not";*

B-30"*A holistic view of the system is essential for identifying the elements of a system as well as its advantages, disadvantages, and bottlenecks*".

Although some statements were not chosen in large numbers, their justifications gain significance when analyzed together. The problem-solving-oriented vision takes on an interesting dimension when we examine the students' claims:

B-17 "*With the notion of the importance of the system's structures, it becomes easier to distinguish what is relevant to the problem to be solved, and whether it is feasible to proceed with the implementation*";

A-22 "*Being able to create design strategies for sustainability is the premiss for all others design work that comes after & it is relevant throughout the whole project process as things change*";

B-26 "*Strategy is one of the most important pillars for solving problems and projects*";

3.2 Collaboration with the people involved provides a deep and authentic understanding of reality.

Collaboration was also strongly present, by recognizing that they need the collaboration of stakeholders to understand their reality and better contextualize the problem. According to them, this recognition of reality is important to facilitate the search for a more efficient solution to the problem, as follows:

A-2 "*Firstly it's necessary to understand the current situation to see a problem's context and work towards solving it - which requires collaboration with actors in this context as they are the ones who have to realize a solution*";

A-26 "*I find working with multiple teams most important, as I think it is important to include different perspectives when solving sustainability problems - I think this strengthens the solution*".

Collaboration can again be interpreted as interdisciplinarity, as analyzed in the statement below:

B-16 "*Working with experts from various disciplines brings different possibilities, creative ideas, and a much better-expected result*";

3.3 Interdisciplinary as 'transversal' and connected competence

Although interdisciplinarity is not determined as a key competence for sustainability (Wiek, 2011), students understood its importance. In their speeches justifying the choice of the most important statements, interdisciplinarity appears integrated with systemic thinking and collaboration skills.

We can notice that, even though students chose statements related to systemic thinking, their justifications point to a view related to interdisciplinarity, as in the following statements:

B-3 "*Complex problems in various disciplines involve deconstructing a large problem into smaller problems, thus enabling them to be solved by different specialists*";

A-4 "*I think that this is our superpower as SDE! To understand complex problems through multiple lenses/disciplines. I think that real sustainable solutions call for multiple disciplines, and I think it is very unique for our disciplines that we can manage this*";

A-21 "*We as humans will never have all the knowledge in our heads as multiple specialists on their jobs. We need to listen to them and gather the all-important data to facilitate sustainability*".

Similarly, collaboration can be interpreted as interdisciplinarity, as analyzed in the statement below:

B-16 "*Working with experts from various disciplines brings different possibilities, creative ideas, and a much better-expected result*";

3.4 Criticisms towards SDGs, radicalism, and lack of flexibility as important reflections when it comes to acting for sustainability

When asked about the least important competences, students refuted the idea of associating projects with the UN SDGs. According to them, the SDGs are general objectives and do not consider the particularities of each region, and their use as a reference can hide important local problems, as in the following comments:

A-3 "*The SDG is very vague. New opportunities could be explored by people/stakeholders on their own since they have an intrinsic desire to push forward the sustainable agenda";*

A-13 *"Because it is incredibly prescriptive, scenarios are excellent tools but are building on limited scenarios, limited views of the world*";

A-23 *"To solve the issues at hand we would need to look at the local problems and environments. That can be very difficult when working with SDGs as many of these are global or at the very least international. It is a problem because it is difficult to fix the world if you don't even know how to fix your country or region first"*.

On the other hand, the word "radical" in one of the statements was interpreted in two ways: radicalism as a violation of democratic values (B-14 "*When you expose your radical personal positions, there is a great risk of generating discord among the group*") or as inflexibility (B-12 "*It is important to have flexibility in the face of challenges and agents*").

Both positions, whether a criticism of the SDGs or a concern with radicalism, regardless of the interpretation, demonstrate an awareness and consideration for contextual elements that determine the problems but also the sustainability of solutions. This means that one solution does not fit all and that even though sustainability challenges are global, collaborations, problems, and solutions need to be addressed with local knowledge and capabilities.

4 Discussion and conclusion

According to the results presented, we realized that students' perceptions regarding the sustainability competences necessary for engineers to solve complex problems related to sustainability are aligned with the definitions of sustainability and the standard of education for sustainability, which are systemic thinking, interdisciplinarity and collaboration (Guerra & Holgaard, 2019; Guerra & Smink, 2019; Sterling, 1996, 2001). These findings suggest that students recognize the need to understand complex and interconnected systems that sustain socio-environmental challenges, in addition to valuing the need to use different experts from different areas to address the problem more comprehensively and effectively.

The justifications also show that interdisciplinarity is perceived as transversal and plays a role in connection with different competences. For example, students stress it as relevant in collaborative processes, whilst in others they connect to systems thinking and complexity.

Taking into account that all key competences for sustainability are equally important and that people, especially, in our case, engineers, need to develop them together, our findings can guide the development of educational strategies to develop these competences and values among students. On the other hand, the convergence in views of students from universities in different countries can be attributed to a combination of factors, including the globalization of knowledge, similarities in engineering curricula, academic and environmental culture, global environmental sensitivity, and similarities in sustainability challenges faced around the world. In this way, a joint, global effort to plan efficient means for developing key competences for sustainability is valid.

In conclusion, the results obtained highlight a multidimensional approach, where systemic thinking, collaboration, interdisciplinarity, and problem-solving are highly recognized and prioritized to be part of engineering education for sustainability action. They recognize the importance of education for sustainability as a boost for future generations, along with the ability to analyze reality and identify challenges, aiming for an equitable and sustainable society. These findings reflect students' holistic and conscious understanding of sustainability, offering valuable guidelines for educational strategies and initiatives aimed at developing these

skills and values among students. In this context, the incorporation of sustainability themes into existing disciplines and the use of active methodologies, such as Problem/Project Based Learning (PBL), can be useful not only develop students' theoretical understanding of sustainability but also encourage the application practice and development of values and sustainable behaviors that will last throughout their lives.

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